



A Triphenylamine-Functionalized Co-Assembly Material for Record Efficiency Inverted CsPbI₃ Perovskite Solar Cells

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Abstract: The precise regulation of self-assembled monolayer (SAM) distribution and interfacial modification is pivotal for advancing the performance of p-i-n inverted perovskite solar cells (PSCs). Here, a new co-assembly material, 4-(aminomethyl)-*N,N*-diphenylaniline iodide (TPAI), is developed to make SAM orderly assembled. Density functional theory (DFT) calculation and sum frequency generation (SFG) spectroscopy reveal that TPAI binds with SAM via π - π interactions, effectively suppressing SAM aggregation and enhancing the orderliness of self-assembly. Further characterization by Fourier transform infrared spectroscopy (FTIR) and X-ray photoelectron spectroscopy (XPS) demonstrates that $-\text{NH}_2$ group in TPAI coordinates with undercoordinated Pb^{2+} to passivate defects of cesium lead triiodide (CsPbI_3) film. The TPAI modification creates a defect-minimized buried interface with optimized energy alignment, significantly improving hole extraction and transport kinetics. Consequently, the TPAI-treated CsPbI_3 PSCs achieve a high power conversion efficiency (PCE) of 21.60%, the highest reported value for inverted CsPbI_3 PSCs, maintaining 96.71% initial PCE after tracking at maximum power point (MPP) for 1400 h. This work provides a molecular-level strategy for interfacial engineering, advancing the development of efficient and durable perovskite photovoltaics.

Introduction

Perovskite solar cells (PSCs) have developed rapidly in recent years, thanks to the rapid improvement of their efficiency and stability. According to the spatial distribution of each functional layer, PSCs can be categorized into inverted (p-i-n) and normal (n-i-p) architectures. Compared with normal PSCs, inverted PSCs exhibit higher stability and power conversion efficiency (PCE), with the current maximum efficiency reaching up to 26.7%.^[1] In addition, inverted PSCs can be applied to construct perovskite/silicon tandem

solar cells with a state-of-the-art efficiency of 34.6%.^[2–6] Nevertheless, further improving the efficiency and stability of perovskite/silicon tandem solar cells still faces several challenges.

The first challenge is how to prepare high-quality self-assembled monolayers (SAMs) for efficient hole transport. Nowadays, SAMs are commonly used as hole transport layers (HTLs) in high-efficiency inverted PSCs, such as [2-(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz), [2-(3,6-Dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz) and (4-(3,6-dimethyl-9H-carbazol-9-yl)butyl) phosphonic acid (Me-4PACz).^[7–10] However, SAMs often tend to aggregate on the substrates, resulting in uneven HTL, which results in current leakage and non-uniform perovskite film, thereby limiting the performance of inverted PSCs.^[11–13]

Many strategies have been investigated to address the issue of non-uniform SAMs. For example, Liu et al. proposed a hybrid SAM strategy by mixing 4,4',4''-nitrotribenzoic acid (NA) and Me-4PACz to disperse SAMs and improve the wettability of the perovskite solution, thus effectively releasing the stress at the buried interface, and realizing a high efficiency of 26.69% for $\text{FA}_{0.95}\text{Cs}_{0.05}\text{PbI}_3$ devices (FA: formamidinium).^[14] Chang et al. proposed a co-assembly strategy to enhance the tridentate anchoring and ordered arrangement of SAMs through π - π interactions and hydrogen bonding between a 2D π -conjugated structured derivative (PAG) and SAMs. The devices based on this strategy finally achieved a remarkable PCE of 26.10%.^[15] Li et al. also proposed a co-adsorb (CA) strategy by introducing 2-chloro-5-(trifluoromethyl)isonicotinic acid (PyCA-3F) between 2PACz and perovskite, and a PCE of over 25% was

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Additional supporting information can be found online in the Supporting Information section

obtained.^[16] Zhang et al. solved the wettability and spreading issues of SAMs by using a co-assembled layer and buffer layer. First, 4-*N,N*-dimethylbenzeneboronic acid hydrochloride (4NPBA) was co-assembled with SAMs and then formamidine iodide (FAI) was utilized as the buffer layer to prepare dense and flat perovskite films, resulting in an exceptionally high fill factor (FF) of 88.35% in the inverted PSCs.^[17]

The second challenge for constructing stable perovskite/silicon tandem PSCs is the unstable factors caused by organic components of hybrid perovskite in the inverted PSCs. Compared with organic-inorganic perovskites, all inorganic cesium lead triiodide (CsPbI₃) perovskite is more stable under illumination and thermal stress, and well suited for tandem solar cells with silicon, thus realizing pure inorganic tandem solar cells.^[18–21] However, there are still few reports on the high-efficiency inverted CsPbI₃ PSCs, mainly due to the wettability and spreading issues of SAMs.

In this work, in order to obtain a high-efficiency inverted CsPbI₃ PSCs, a new co-assembly material, 4-(aminomethyl)-*N,N*-diphenylaniline iodide (TPAI), is specifically developed to make the SAM orderly assembly. Density functional theory (DFT) calculation and sum frequency generation (SFG) analysis reveal that TPAI can bind with the SAM through π - π interactions to effectively improve the orderly assembly of the SAM on the surface. In addition, Fourier transform infrared spectroscopy (FTIR), X-ray photoelectron spectroscopy (XPS), and scanning electron microscopy (SEM) characterizations confirm that the co-assembly TPAI could bind with upward perovskite by $-\text{NH}_3$ and Pb interaction, which is beneficial for spreading the perovskite solution and thus improving the quality of the perovskite film. Consequently, the TPAI-treated devices achieve a high open-circuit voltage (V_{OC}) of 1.26 V, and a PCE of 21.60%, which is the highest reported PCE for inverted CsPbI₃ PSCs. Additionally, the TPAI-treated device exhibits respectable operational stability, maintaining 96.71% of the initial PCE after tracking at maximum power point (MPP) for 1400 h.

Results and Discussion

The synthesis route and characterizations of TPAI are shown in Scheme S1 and Figure S1–S3 (Supporting Information). Ethanol solutions of TPAI with different concentrations were spin-coated onto nickel oxide (NiO_x)/SAM (2PACz) substrates, followed by annealing at 100 °C for 10 minutes, to regulate the distribution of the SAM on the substrates. The detailed synthesis of TPAI is described in the Supporting Information. The effect of TPAI treatment on the surface morphology of NiO_x/SAM was examined by scanning electron microscope (SEM). As shown in Figure 1a,b and Figure S4 (Supporting Information), compared with the control NiO_x/SAM substrate without the TPAI treatment, the TPAI-treated substrates exhibit different surface morphology and hydrophobicity. As the treatment concentration of TPAI increases, the coverage of SAM on the substrates gradually increases, indicative of a reduction in dark pores. The surface of substrates becomes smoother and more hydrophilic, implying that TPAI could probably induce the rearrangement

of SAM and reduce surface tension. For the TPAI-treated substrates, the improved SAM coverage and reduced surface tension is beneficial for the coverage of perovskite films and for blocking the transmission of electrons. When the perovskite solution was dropped onto the substrates, the perovskite solution exhibited better wettability and faster-spreading speed on the NiO_x/SAM/TPAI substrates than on the control substrate (Figure S5, Supporting Information).

The influence of the TPAI treatment on the surface of NiO_x/SAM substrates was also characterized by atomic force microscopy (AFM). As shown in Figure 1c,d, the surface roughness (R_q) of NiO_x/SAM/TPAI was decreased compared with that of NiO_x/SAM. The lower roughness and smaller water contact angle of the treated substrate is more favorable for the deposition of perovskite film. The surface contact potential difference (CPD) of NiO_x/SAM and NiO_x/SAM/TPAI substrates was tested by Kelvin probe force microscopy (KPFM) (Figure S6, Supporting Information). The CPD was significantly improved from 171 to 263 mV. A higher CPD means an upward Fermi level (E_{Fermi}) energy level for the treated NiO_x/SAM substrate, which is consistent with the results of the ultraviolet photoelectron spectroscopy (UPS) (Figure S7 and Table S1, Supporting Information).^[22,23]

To illustrate the effect of TPAI-treatment on the upper perovskite films, the top-view SEM of CsPbI₃ perovskite film was tested. As shown in Figure 1e,f and Figure S8, Supporting Information, the grain size of perovskite film has no significant change before and after TPAI-treatment. The water contact angle was increased after TPAI treatment due to the smoother and less porous perovskite films, which is expected to be helpful for the stability of perovskite films. The AFM images of perovskite films with and without TPAI treatment are shown in Figure 1g,h and Figure S9 (Supporting Information). It can be seen that the surface roughness of the perovskite films with treatment was reduced compared with the film without TPAI treatment, the perovskite film on 3 mg mL⁻¹ TPAI-treated substrate shows the lowest surface roughness. 3 mg mL⁻¹ TPAI is used as an optimal treatment concentration in the following studies unless otherwise stated.

The crystallinity of perovskite films was studied by X-ray diffraction (XRD). As shown in Figure 1i and Figure S10 (Supporting Information), the perovskite films exhibit diffraction peaks at 13.80° and 28.30°, corresponding to the (110) and (220) crystal planes of CsPbI₃. The TPA-treated perovskite film exhibited higher diffraction peaks, indicating an improved preferred growth orientation of perovskites.^[24,25] As shown in Figure 1j and Figure S11a (Supporting Information), the absorption spectra of CsPbI₃ films were tested by ultraviolet-visible (UV-Vis) absorption spectroscopy. Both films show similar absorption edges at 730 nm, corresponding to a bandgap of 1.7 eV (Figure S11b in Supporting Information).^[26] In the 500–700 nm wavelength range, there is a slight increase in the absorption intensity of the TPAI-treated film, which can be attributed to the crystallinity improvement of the treated film and is consistent with the XRD results.

The effect of TPAI treatment on the charge transfer at the NiO_x/SAM/CsPbI₃ interface was assessed by the steady-state

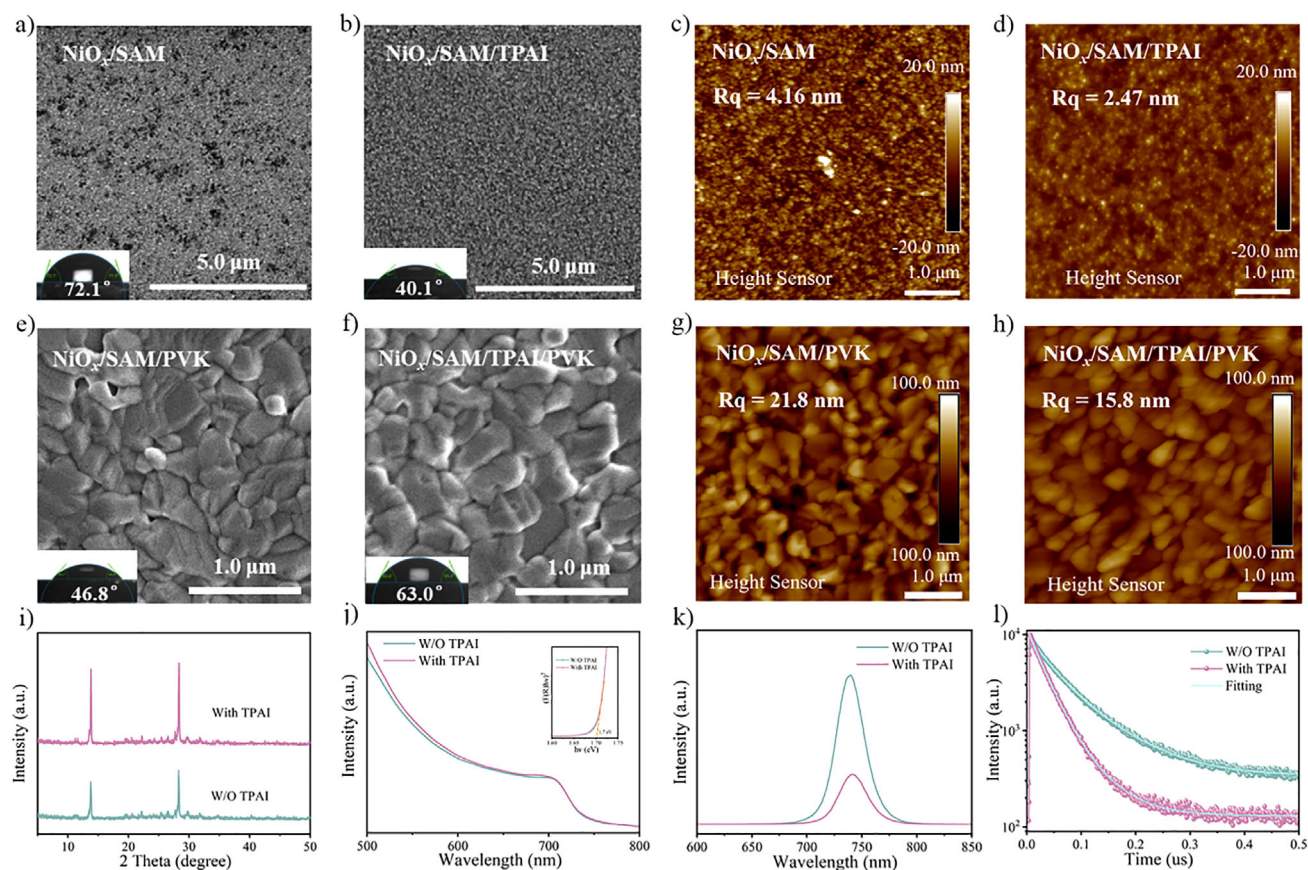


Figure 1. Top-view SEM images and contact angles of a) NiO_x/SAM , b) $\text{NiO}_x/\text{SAM}/\text{TPAI}$, AFM images of c) NiO_x/SAM , d) $\text{NiO}_x/\text{SAM}/\text{TPAI}$. SEM images of e) $\text{NiO}_x/\text{SAM}/\text{PVK}$ and f) $\text{NiO}_x/\text{SAM}/\text{TPAI}/\text{PVK}$ films. AFM images of g) $\text{NiO}_x/\text{SAM}/\text{PVK}$ and h) $\text{NiO}_x/\text{SAM}/\text{TPAI}/\text{PVK}$ films. i) XRD patterns, j) UV-Vis absorption spectra and tauc spectra, k) The steady-state PL spectra, and l) TRPL curves for $\text{NiO}_x/\text{SAM}/\text{PVK}$ and $\text{NiO}_x/\text{SAM}/\text{TPAI}/\text{PVK}$ films.

photoluminescence (PL) and time-resolved photoluminescence (TRPL). As shown in Figure 1k and Figure S12a (Supporting Information), the intensity of the PL peaks was significantly decreased after TPAI treatment, indicating an improved charge transfer between TPAI-treated NiO_x/SAM and CsPbI_3 . The TRPL spectra are fitted by the doubled-exponential decay function, and the results and the fitted parameters are presented in Figure 1l, Figure S12b and Table S2 (Supporting Information). The TPAI-treated hybrid film shows a shorter average carrier lifetime (38.31 ns) than that of the control hybrid film (72.45 ns), which is consistent with the PL tendency. The PL and TRPL results indicate that TPAI treatment facilitates the hole transfer from perovskite to SAM.^[27]

In order to investigate the influence of TPAI treatment on the buried surface of perovskite film, the perovskite film was stripped from the (indium tin oxide) ITO/ NiO_x/SAM substrate by ultraviolet (UV) curing glue.^[28,29] As shown in Figure 2a,b, the surface of the SAM and the buried surface of the perovskite film are fully exposed after stripping. According to the elemental mapping results (Figures S13 and S14, Supporting Information), no residual perovskite is tested on the peeled ITO/ NiO_x/SAM substrate (region I), and the SAM still remained on substrates (Figure 2c,d). The top-

view morphology of the buried surface of perovskite films (region II) was observed by SEM.^[30] As shown in Figure 2e,f, compared with the control perovskite film, the TPAI-treated film exhibits smoother and denser surface morphology and a more uniform grain size distribution. As shown in Figure 2g,h, the cross-sectional SEM images confirm the surface results, that is, the TPAI-treated perovskite film has a smoother surface and larger grains with compact arrangement relative to the control film.^[31]

The surface potential of the buried perovskite surface was evaluated by KPFM. As shown in Figure 2i,j, the CPD value of perovskite was increased from 53 to 110 mV and became more uniform after TPAI treatment. The enhanced CPD value corresponds to a higher conduction band (E_c) and E_{Fermi} for TPAI-treated film, which is confirmed by the UPS results (Figure S15 and Table S3, Supporting Information). The E_c and E_{Fermi} of the perovskite film were elevated from -3.56 to -3.53 eV and -5.05 to -4.75 eV after TPAI treatment, respectively. The optimized surface of perovskite film is more conducive to blocking electrons and transporting holes at the buried interface.^[32]

The effect of TPAI treatment on the residual crystal strain was examined by grazing-incidence X-ray diffraction (GIXRD). As shown in Figure 2k,l, with the increase of

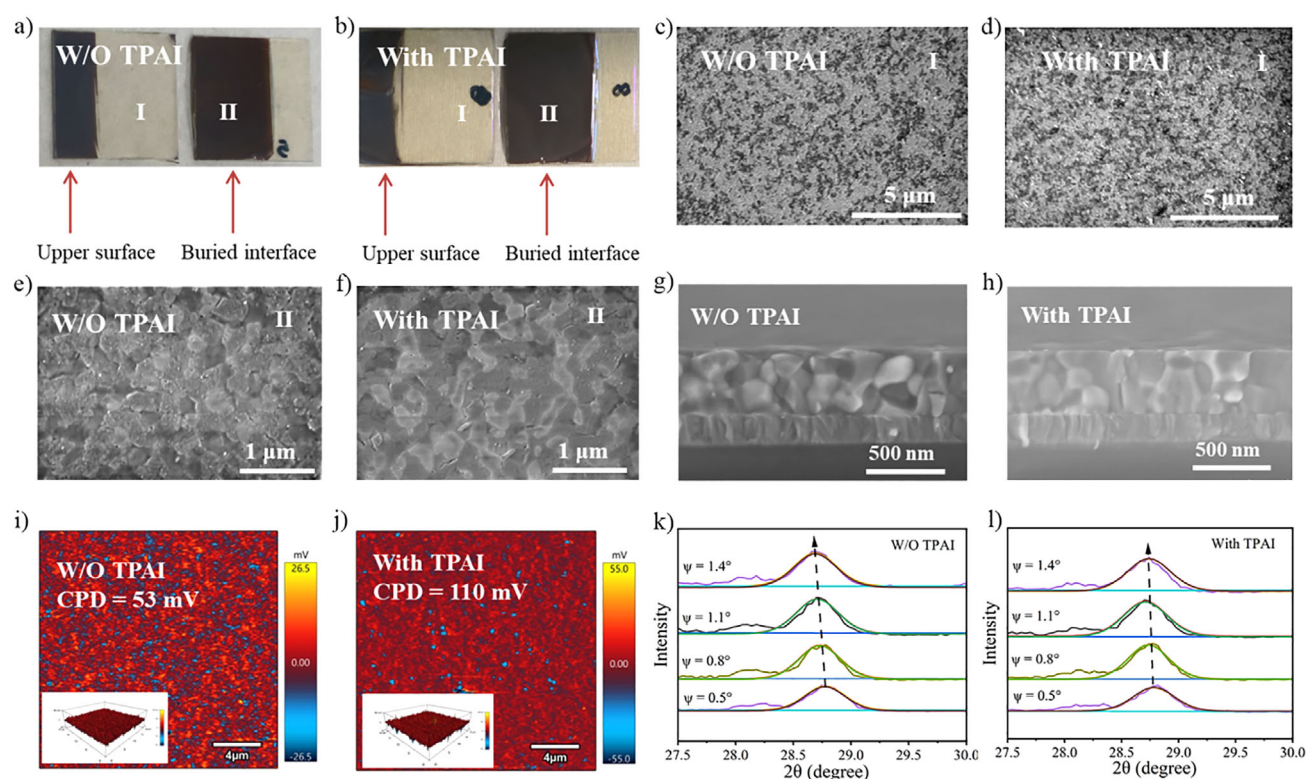


Figure 2. The photo of perovskite film buried interface without a) and with b) TPAI treatment. Top-view SEM images of residual area (I) after peeling of perovskite films without c) and with d) TPAI treatment. Top-view SEM images of perovskite film buried interface (II) without e) and with f) TPAI treatment. Cross-sectional SEM images of g) ITO/NiO_x/SAM/CsPbI₃/PC₆₁BM/BCP/Ag and h) ITO/NiO_x/SAM/TPAI/CsPbI₃/PC₆₁BM/BCP/Ag. KPFM images for perovskite film buried interface without i) and with j) TPAI treatment. GIXRD spectra of without k) TPAI and with l) TPAI treated CsPbI₃ films at different slant angles.

incident angle (Ψ) from 0.5° to 1.4°, the diffraction peak corresponding to the (220) plane of the control CsPbI₃ film gradually shifted toward the lower 2θ angle, a signature of residual tensile strain presents within the control film. In contrast, the film with TPAI treatment alleviates residual tensile strain, as evidenced by the almost unshifted diffraction peaks.^[33,34]

The electron-rich characteristic of triphenylamine (TPA) in TPAI is expected to have a strong π - π interaction with the carbazole moiety in the SAM. To verify this hypothesis, FTIR spectroscopy was conducted to gain insights into the intermolecular interactions between TPAI and 2PACz. As shown in Figure 3a, the characteristic peak of the benzene ring in 2PACz at 1526.69 cm⁻¹ was shifted to a higher wavenumber by 11.03 cm⁻¹ upon mixing 2PACz with TPAI. This shift can be attributed to π - π interaction between the SAM and TPAI.^[35] As shown in Figure 3b,c, 2PACz exhibits two absorption bands at 1011.15 and 943.76 cm⁻¹, respectively, corresponding to the P—OH stretching vibration, and two absorption bands at 1241.51 and 1180.47 cm⁻¹ associated with the vibration of P=O. The shifts of absorption bands for P—OH and P=O are 12.35 and 13.69 cm⁻¹ after mixing 2PACz and TPAI, respectively. The FTIR results reveal that there is a strong interaction between 2PACz and TPAI.^[36,37]

XPS was further performed to verify the interaction between 2PACz and TPAI. As shown in Figure 3d, the P 2p peak of the phosphate group for 2PACz shifted significantly

toward lower binding energy after mixing 2PACz and TPAI. The O 1 s peaks of a phosphate group, corresponding to P—OH and P=O, also shifted to lower binding energies (Figure 3e).^[38] The shifts of P 2p and O 1 s peaks of 2PACz further confirms the interaction between 2PACz and TPAI, which is in good agreement with FTIR results. As depicted in Figure 3f, the peak of N 1 s corresponding to 2PACz is located at 400.30 eV, and TPAI has two peaks of N at 401.85 and 400.01 eV due to the presence of two forms of N. When TPAI was mixed with 2PACz, the N 1 s peak of TPAI shifted to higher binding energy.

Moreover, the FTIR and XPS were also performed to elucidate the interaction between TPAI and perovskite. As shown in Figure 3g, the stretching peak of —NH₃ of TPAI is shifted to a lower wavenumber when mixing TPAI with CsI or PbI₂. XPS results indicate that after mixing PbI₂ with TPAI, the binding energy of Pb element is shifted upward (Figure 3h), and the binding energies of both the I and N elements are shifted upward (Figure 3i and Figure S16, Supporting Information), which proves that the empty orbital of Pb²⁺ can establish coordination bonds with the lone pair electrons of —NH₃ in TPAI.^[35,39] After CsI is mixed with TPAI, the binding energies of Cs, I, and N elements are also shifted toward higher binding energies, suggesting that there is also a strong interaction with CsI, which is consistent with the FTIR results (Figure 3j,k; Figure S16, Supporting Information). To qualitatively study the deposition change

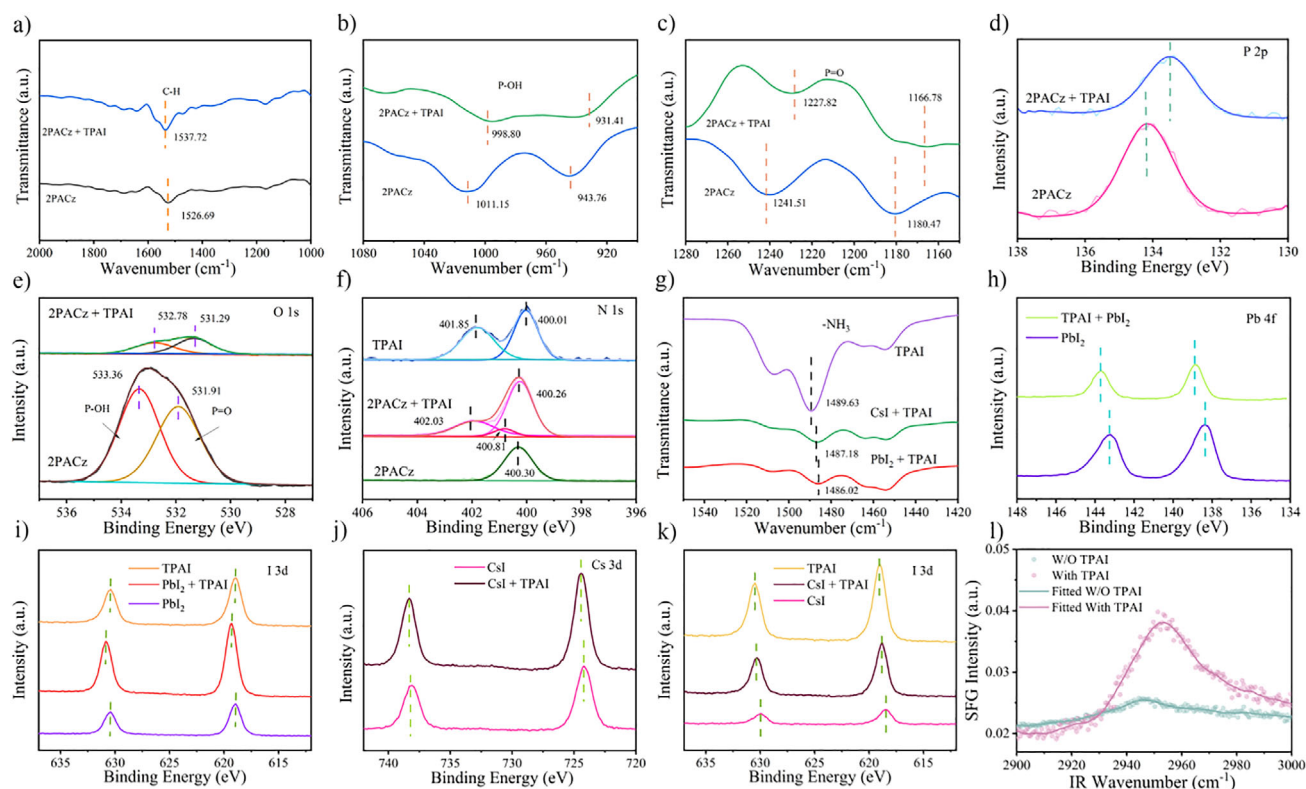


Figure 3. a) FTIR spectra of 2PACz and 2PACz + TPAI films. b) The P-OH and c) P = O FTIR spectra for 2PACz and 2PACz + TPAI films. XPS spectra of d) P 2p and e) O 1s for 2PACz and 2PACz + TPAI films. f) XPS spectra of N 1s for 2PACz, 2PACz + TPAI and TPAI films. g) FTIR spectra of TPAI, CsI + TPAI and PbI₂ + TPAI films. h) The XPS spectra of Pb 4f for TPAI and PbI₂ + TPAI films. i) The XPS spectra of I 3d for TPAI, PbI₂ and PbI₂ + TPAI films. j) The XPS spectra of Cs 3d for CsI and CsI + TPAI films. k) The XPS spectra of I 3d for TPAI, CsI, and CsI + TPAI films. l) SFG ssp spectra for the control and TPAI-treated Glass/NiO_x/SAM films.

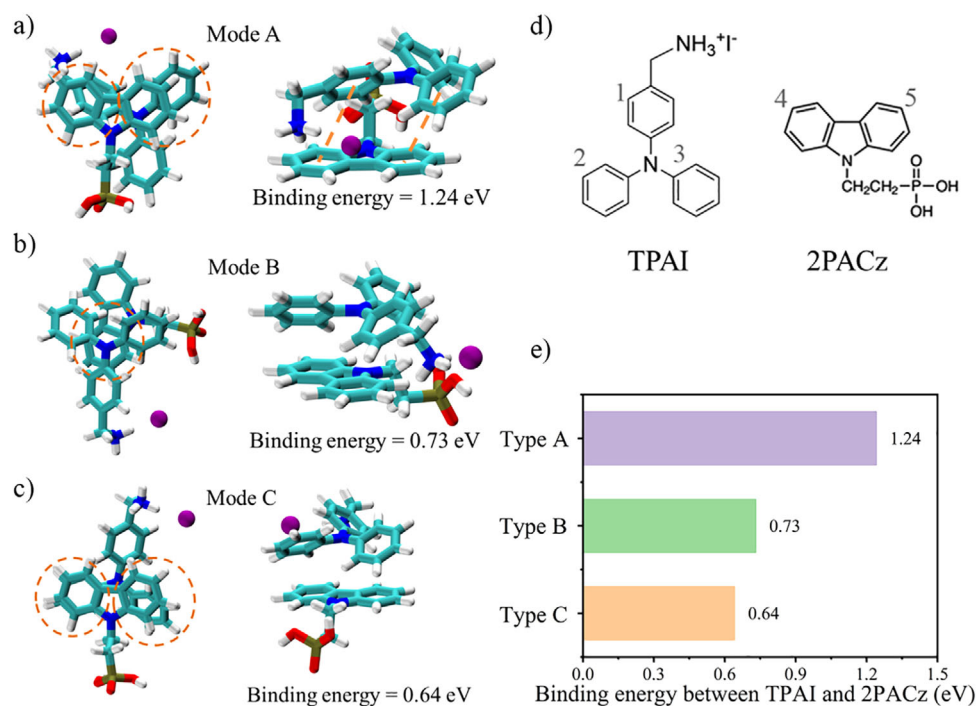


Figure 4. a–c) The main and top views of three different DFT calculation models of 2PACz and TPAI. d) Planar structural formulations of 2PACz and TPAI. e) The calculated binding energy between 2PACz and TPAI.

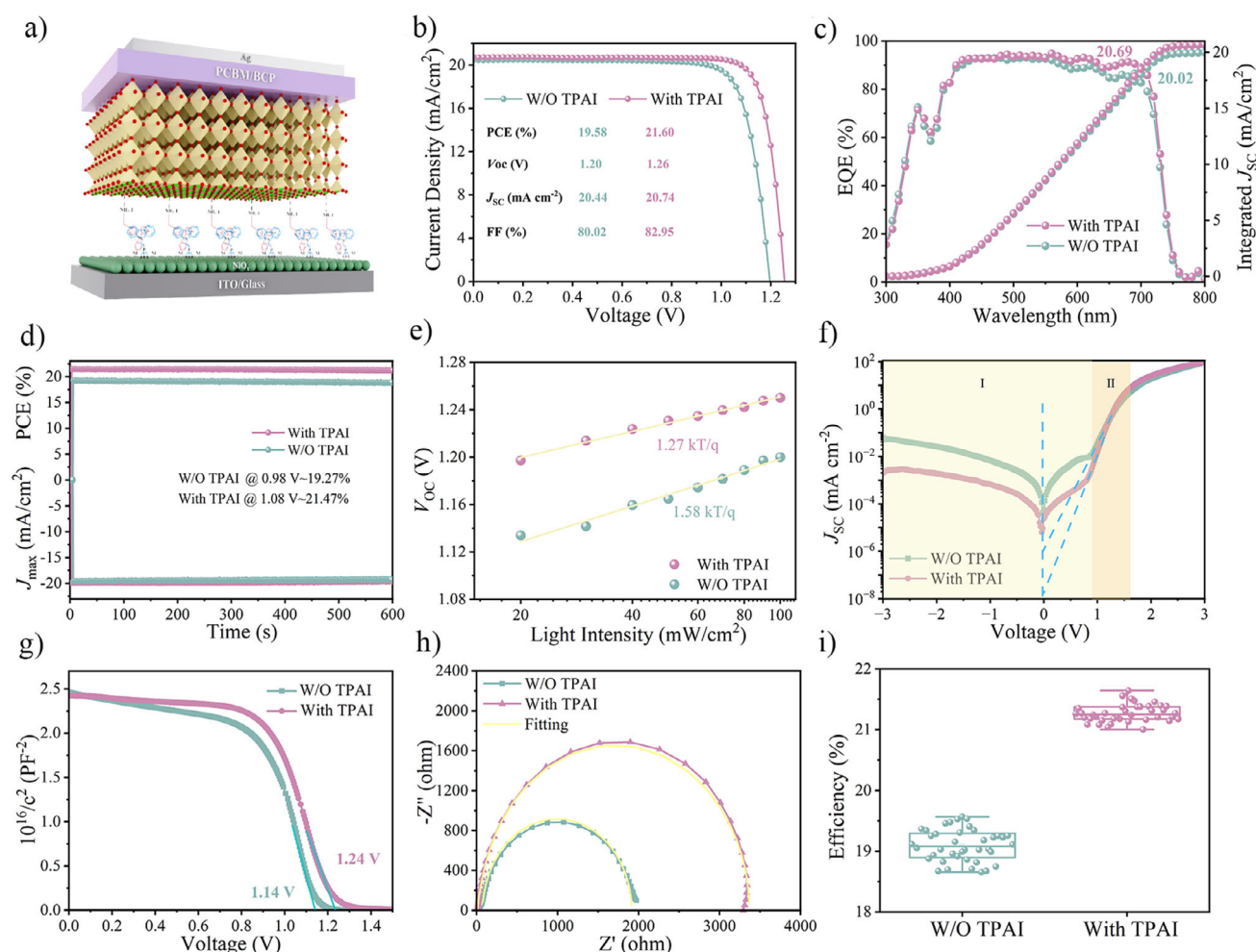


Figure 5. a) Device structure of ITO/NiO_x/SAM/TPAI/CsPbI₃/PC₆₁BM/BCP/Ag. b) J-V curves, c) EQE curves, d) the stable output curves, e) V_{oc} versus light intensity f) the current versus voltage in dark conditions, g) the C-V curves, and h) Nyquist plots of the inverted CsPbI₃ PSCs without and with TPAI modification, respectively. i) The statistic distributions of PCE for 40 inverted CsPbI₃ PSCs without and with TPAI treatment.

of the SAM on the substrate after TPAI treatment, the SFG vibrational spectroscopy was applied, which is a powerful second-order nonlinear optical technique for selectively exploring molecules at surface and buried interfaces.^[40,41] As shown in Figure 3l, the vibration peak of C–H group at $\approx 2950\text{ cm}^{-1}$ was detected in the pure SAM film, which was enhanced after TPAI treatment, and attributed to the improved molecular orientation of SAM.^[42]

DFT calculation was conducted to further understand the interaction between TPAI and 2PACz. Figure 4a–c shows the interaction mode between 2PACz and TPAI and the corresponding interaction energies. Figure 4d shows the molecular structures of TPAI and 2PACz, in which the benzene rings are labeled 1–5 to better illustrate their interactions. In the Mode A, the benzene rings 1, 2 in TPAI interact with benzene rings 4, 5 in 2PACz; In the Mode B, the center of TPAI interacted with carbazole in 2PACz; In the Mode C, the benzene rings 2, 3 in TPAI interact with benzene rings 4, 5 in 2PACz. The binding energy of Mode A is 1.24 eV, which is the highest value of the three modes (Figure 4e). In this binding mode, the downward phosphonic acid can anchor to NiO_x,

and the upward –NH₃ group can be bound with perovskite. Therefore, the TPAI plays a bridging role in enhancing the orderly arrangement of the SAM through π - π interactions and improving the quality of the buried surface of perovskite through the interaction with PbI₂ and CsI, which is conducive to the photovoltaic performance of PSCs.

To evaluate the effect of TPAI treatment on the photovoltaic performance, inverted CsPbI₃ PSCs were fabricated with a configuration of ITO/NiO_x/SAM/TPAI/CsPbI₃/[6,6]-phenyl-C₆₁-butyric acid methyl ester (PC₆₁BM)/bathocuproine (BCP)/Ag (Figure 5a). Figure 5b and Figure S17 (Supporting Information) shows the current-voltage (J-V) curves of the PSCs. The control device shows a maximum PCE of 19.58%, a V_{oc} of 1.20 V, a short-circuit current density (J_{sc}) of 20.44 mA cm⁻², and an FF of 80.02%. In contrast, the TPAI-treated PSC exhibits a substantially improved PCE of 21.60%, with a V_{oc} of 1.26 V, a J_{sc} of 20.74 mA cm⁻², and an FF of 82.95% (Tables S4 and S5, Supporting Information). According to the external quantum efficiency (EQE) curves, the integrated J_{sc} values of the control and TPAI-treated PSCs are 20.02 and 20.69 mA cm⁻²,

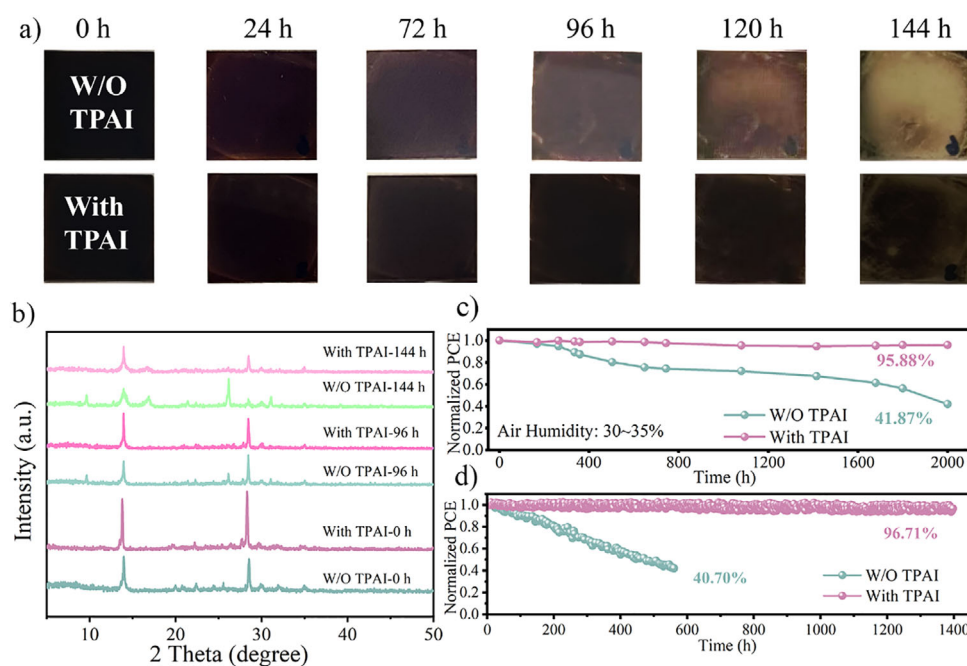


Figure 6. a) The photographs for CsPbI₃ films without and with TPAI modification aging for different times at 35% RH. b) XRD patterns of CsPbI₃ films without and with TPAI treatment at 35% RH for 0–144 h. c) The air stability of unencapsulated PSCs without and with TPAI modification after 2000 h. d) The MPP tracking curves of encapsulated PSCs without and with TPAI modification under continuous light illumination (100 mW cm⁻²) with ≈30% RH for 1400 h.

respectively (Figure 5c), which is consistent with the results derived from J - V curves.^[43] The improved performance of TPAI-treated devices could be attributed to the optimized energy level and improved hole extraction at the buried interface. The stabilized current density and efficiency of the PSCs were tracked at the MPP for 600 seconds. As shown in Figure 5d, the stabilized PCE of the TPAI-treated device is 21.47% at 1.08 V bias, which is higher than that of the control device (19.27%, 0.98 V bias) and agrees with the values derived from the J - V curves. The dependence of V_{OC} on light intensity was measured to reveal the charge recombination mechanism. As shown in Figure 5e, the ideality factors (n) of control and TPAI-treated PSCs were calculated to be 1.58 and 1.27 $k_B T q^{-1}$, respectively.^[44] The ideal factor of the TPAI-treated PSC is closer to 1, suggesting reduced trap-assisted recombination in the TPAI-treated device. The trap-state densities (N_{trap}) of the devices were determined by the space-charge limited current (SCLC) method. As shown in Figure S18 (Supporting Information), the N_{trap} was significantly decreased from 4.01×10^{15} to 1.63×10^{15} cm⁻³ after TPAI treatment, indicating that TPAI can effectively inhibit the defect formation in CsPbI₃ films.^[45,46]

The dark-state J - V and the capacitance-voltage (C - V) curves were tested to explore the reason for the improvement in V_{OC} after TPAI treatment. As shown in Figure 5f, the reverse leakage current of the TPAI-treated device is lower than that of the control device in region I, indicating a suppressed charge recombination. In Region II, the intercept of the exponential range linear fitted at 0 V corresponds to the reverse saturation current (J_0). The J_0 of the TPAI-treated device is lower than that of the control device,

based on the relationship between V_{OC} and J_0 , the V_{OC} of the TPAI-treated device should be higher than that of the control device, which is in better agreement with the J - V curve.^[47] The built-in voltage (V_{bi}) of devices was tested by the Mott-Schottky experiment (Figure 5g). The TPAI-treated device exhibits a higher V_{bi} of 1.24 V, than the control device (1.14 V), suggesting a stronger driving force for photo-generated charge separation and transfer in the TPAI-treated device, contributing to the enhancement of V_{OC} .^[48,49]

To explore the reason for the enhancement of FF, the electrochemical impedance spectroscopy (EIS) was implemented (Figure 5h), and the equivalent circuit model and the corresponding fitted parameters are displayed in Figure S19 and Table S6 (Supporting Information). Compared with the control device ($R_{rec} = 1835.00 \Omega$, $R_s = 25.98 \Omega$), the TPAI-treated device shows a larger recombination resistance ($R_{rec} = 3311.00 \Omega$) and a smaller series resistance ($R_s = 14.66 \Omega$), suggesting that the TPAI treated interface could improve charge transfer and suppress carrier recombination to promote FF.^[50,51] Additionally, by counting 40 independent devices for each inverted CsPbI₃ PSCs, the average PCE of the target devices is 21.27%, and higher than 19.07% of the control devices (Figure 5i), indicating an excellent reproducibility of the TPAI inverted device. The statistics of V_{OC} , FF, and J_{SC} of devices are shown in Figure S20 (Supporting Information).

The stability of perovskite films and devices was investigated under various aging conditions. As shown in Figure 6a, the storage stability of CsPbI₃ films was tested in ambient air at a relative humidity (RH) of 35%. The TPAI-treated film retained black after 144 h, while the control film faded

to reddish brown after 96 h, and to yellowish-brown after 144 h. The above aging processes were also monitored by XRD (Figure 6b). The degradation of the TPAI-treated film only manifested as a decrease in peak intensity of α -phase, while the control film showed a severe phase transition from α -phase to δ -phase.

The storage stability of the unencapsulated device was evaluated by storage in air (RH 35% and 25 °C, Figure 6c). The TPAI-treated device retained 95.88% of its initial PCE after 2000 h, while the control device maintained only 41.87% of its initial PCE during the same period. The operational stability of devices was tracked over time in the air at the maximum power point (MPP) under continuous illumination. As shown in Figure 6d, the TPAI-treated device retained 96.71% of its initial PCE after 2000 h illumination, whereas the PCE of the control device rapidly dropped to 40.70% of its original efficiency after only 550 h.

Conclusion

In summary, a novel co-assembly material TPAI was synthesized to improve the quality of the buried interface in inverted CsPbI₃ PSCs through its π - π interaction with the SAM and N-Pb interaction with perovskite. Detailed calculation and experimental results indicated that TPAI treatment diminished SAM aggregation and made SAM orderly assembled, which could enhance surface smoothness and increase the work function of the SAM layer. In addition, the TPAI could bind with upward perovskite by $-\text{NH}_3$ and Pb interaction to spread the perovskite solution, improve the crystallinity, and minimize trap states of perovskite film. So TPAI optimized the quality of the buried interface that augmented hole extraction and transfer capability. Consequently, the TPAI-treated inverted CsPbI₃ PSCs achieved an impressive PCE of 21.60%, along with excellent operational stability. This work not only provides an effective strategy for the development of efficient inverted all-inorganic PSCs, but more importantly, offers potential applications in inorganic perovskite/silicon tandem devices.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords: CsPbI₃ • Co-assembly material • Inverted perovskite solar cells • Self-assembled molecules • Triphenylamine

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